

Sanidhya Vats

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EXPERIENCE

AI Engineer

Jan. 2026 – Mar. 2026

Square Educations

NCR, India

- Built an LLM recommendation engine using NLP classifiers on aptitude/reasoning data to generate tailored career trajectories.
- Established CI/CD deployment and automated Pytest pipelines, boosting code coverage from 65% to 85% and cutting deployment time by 20%.

Growth & Operations Engineer

Aug. 2026 – Present

Geek Room

Delhi, India

- Managed the expansion of the community network, coordinating between multiple regional groups to help them grow their member bases.
- Built and maintained event websites and registration flows that converted outreach into signups.

PROJECTS

Harvest – Autonomous Coding Agent Harness | *TypeScript, Rust, Bun, N-API, Git*

GitHub

- Engineered deterministic execution grounding with pre-read mutation shields, intercepting agent completion turns to enforce test execution against modified files and validate physical Git HEAD progression.
- Built a context compaction engine paired with an in-memory BM25 symbol index, compressing conversational trajectories while retaining active diffs, verification records, and AST symbols without external embeddings.
- Implemented Shannon entropy-gated intent routing to resolve prompt ambiguity via interactive user disambiguation, combined with a native Rust N-API engine for streaming fuzzy patch application.

Oasis – IaC Mutation Testing Framework | *Python, Terraform, OpenTofu, Docker, GitHub Actions*

GitHub

- Designed and engineered an open-source mutation testing framework evaluating the semantic quality of Terraform and OpenTofu test assertions by injecting synthetic faults (mutants).
- Developed an AST parsing engine for IaC configurations along with 12 specialized mutation operators to simulate infrastructure drifts and state anomalies.
- Implemented a zero-drift Git state restoration mechanism ensuring clean workspace recovery post-run, and packaged the tool with self-updating native CLI wrappers.

Real-Time Data Pipeline | *Python, Kafka, FastAPI, Docker*

GitHub

- Engineered an LLM-powered streaming data pipeline using Apache Kafka and Python to ingest real-time data from 3+ external sources, increasing processing throughput by 40% via automated Groq API LLM summarization.
- Enabled RAG-based semantic search by integrating Qdrant (Vector DB) with a resilient web-scraping fallback, containerizing the entire ML inference system with Docker and Kubernetes.

POSITION OF RESPONSIBILITY

Leadership Roles | *Google Developer Groups, Geek Room, ISTE MSIT*

Aug. 2023 – Present

- Mentored 50+ students in AI and full-stack development, moderated 5+ hackathons, and scaled technical talent pools by interviewing and onboarding 250+ new members across all societies.
- Delivered user-friendly websites for community events and organized an Ideathon featuring a Microsoft guest speaker with 100+ participants.

ACHIEVEMENTS

- **Mentored Events:** HackAvensis 2026, Innovortex 3.0 2025, Innerve Hackathon 2026, SIH 2026 Internal Round.
- **Organized Events:** AI Oriented BatterySmart Hackathon 2026, Hack GeekRoom 2026, CodeKshetra 2.0, Trackshift Hackathon 2026, Build-UP Ideathon 2026, Code-Cubicle Hackthon Series, Kaggle-Days Meetup 2025.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++

AI & Machine Learning: PyTorch, Scikit-Learn, Transformers, Hugging Face, LangChain, LangGraph, LLMs, RAG, Prompt Engineering, Vector Databases, Ensemble Learning

Backend & Data: FastAPI, Node.js, Express.js, Flask, Apache Kafka, PostgreSQL, MongoDB, MySQL, Redis, Qdrant, Prisma ORM, Firebase

Frontend: Next.js, React.js, Tailwind CSS, Redux.js, RTK Query, Framer Motion

MLOps & Tools: Docker, Kubernetes, Git, GitHub, Vite, Postman, CI/CD